Glossary

- ODE ordinary differential equation
- PDE partial differential equation
- IVP initial value problem, i.e. a differential equation plus initial conditions that the solution must satisfy
- BVP boundary value problem, i.e. a differential equation plus boundary conditions that the solution must satisfy
- $\mathbb{R}$ set of real numbers
- $\mathbb{R}_+$ set of positive real numbers
- $\mathbb{R}_*$ set of non-negative real numbers
- $\mathbb{R}^k$ generic k-dimensional Euclidean space
- $\mathbb{C}$ set of complex numbers
- $\mathbb{N}$ set of natural (positive integer) numbers, same as $\mathbb{Z}_+$
- $\mathbb{Z}$ set of integer numbers
- $\mathbb{Z}_+$ set of positive integer numbers, same as $\mathbb{N}$
- $\mathbb{Z}_*$ set of non-negative integer numbers
- $x \in \mathbb{R}$ reads "x belongs to $\mathbb{R}$", i.e. x is a real number
- $A \subset B$ reads "A is a subset of B", i.e. any object that belongs to set A also belongs to set B
- $A \cup B$ then union of sets A and B, i.e. the set of elements that belong either to A or to B or to both
- $A \cap B$ the intersection of sets A and B, i.e. the set of elements that belong both to A and B
- $A \setminus B$ the set of elements that belong to A but do not belong to B
- $A \times B$ the set of ordered pairs (a, b) such that $a \in A$ and $b \in B$
- $\mathrm{Re}\,(\cdot)$ real part of
- $\mathrm{Im}\,(\cdot)$ imaginary part of
- a a scalar (an integer, real or complex number)
- $\bar{a}$ the complex conjugate of the scalar a
- $\boldsymbol{a}$ a vector
- $\mathbf{0}$ the vector whose elements are all 0
- $\boldsymbol{A}$ an $n \times m$ matrix with entries a_{ij}
- $\boldsymbol{A}^T$ the transpose of the matrix $\boldsymbol{A}$
- $\bar{\boldsymbol{A}}$ the complex conjugate of the matrix $\boldsymbol{A}$

- $\bar{\boldsymbol{A}}^T$ the adjoint of the matrix $\boldsymbol{A}$
- $\boldsymbol{I}$ the identity matrix
- $\boldsymbol{A}^{-1}$ the inverse of the matrix $\boldsymbol{A}$
- $|\boldsymbol{A}|$ the determinant of the matrix $\boldsymbol{A}$
- $tr(\boldsymbol{A})$ the trace of the matrix $\boldsymbol{A}$
- $|x|$ for $x \in \mathbb{R}$, this is the absolute value: $|x| = \begin{cases} -x, & x < 0, \\ 0, & x = 0, \\ x, & x > 0. \end{cases}$
- $|z|$ for $z \in \mathbb{C}$, this is the modulus: $|z| = \left(\mathrm{Re}\,(z)^2 + \mathrm{Im}\,(z)^2\right)^{1/2}$
- (a, b) open interval between a and b: we write $x \in (a, b)$ when we mean $a < x < b$
- $[a, b]$ closed interval between a and b: we write $x \in [a, b]$ when we mean $a \leq x \leq b$
- $(a, b]$ semi-open interval: $x \in (a, b]$ means $a < x \leq b$
- $[a, b)$ semi-open interval: $x \in [a, b)$ means $a \leq x < b$
- $\forall A$ "for all A, the following (or the preceding) is true"
- $\exists A$ "there exists A such that the following is true"
- $\emptyset$ empty set, i.e. the set containing no elements
- d ordinary differential
- ∂ partial differential
- e base of natural logarithms, $2.718281828459\ldots$
- i imaginary unit, $i^2 = -1$.
- sgn the sign function, defined by $\mathrm{sgn}(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0\,. \end{cases}$